



ASIAN INTERNATIONAL SCHOOL
PRE-BOARD EXAMINATION - II
SUBJECT - SCIENCE
SESSION - 2025-26

MAXIMUM MARKS:80

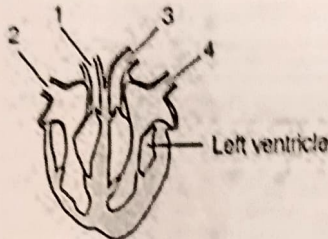
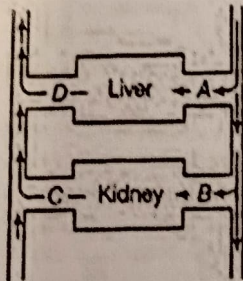
CLASS:X

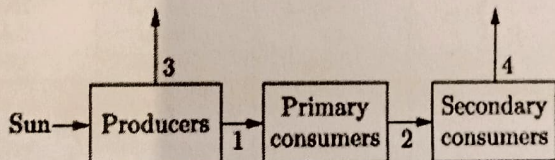
TIME:3 HRS

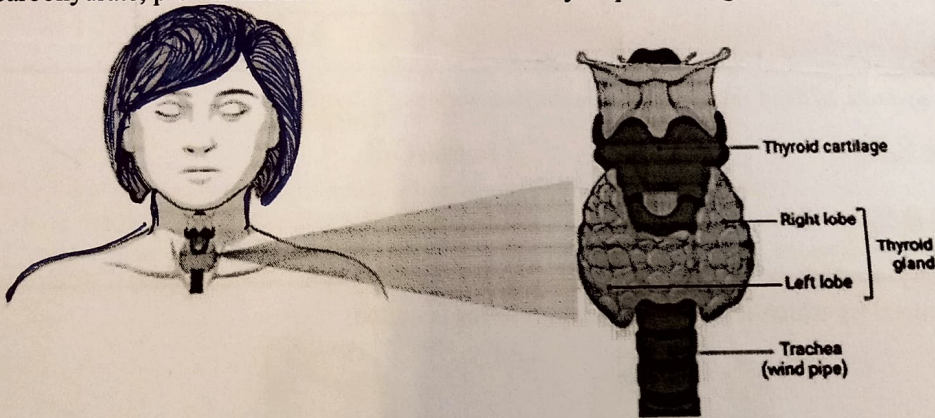
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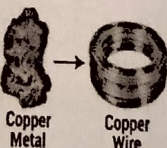
GENERAL INSTRUCTIONS:

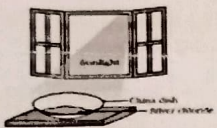
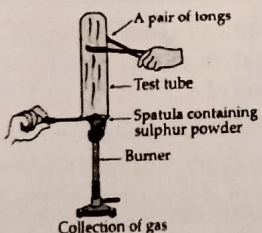
- (i) This question paper consists of 39 questions in 3 sections. Section A is Biology, Section B is Chemistry and Section C is Physics.
(ii) All questions are compulsory. However, an internal choice is provided in some questions. A student is expected to attempt only one of these questions.

S.NO	SECTION A	MARKS																									
1	The diagram shows a vertical section through the heart.  What are the functions of the numbered blood vessels? <table><tr><th></th><th>Carries blood of body</th><th>Carries blood to lungs</th><th>Carries blood from lungs</th><th>Carries blood from body</th></tr><tr><td>(a)</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>(b)</td><td>1</td><td>3</td><td>4</td><td>2</td></tr><tr><td>(c)</td><td>2</td><td>4</td><td>3</td><td>1</td></tr><tr><td>(d)</td><td>3</td><td>1</td><td>4</td><td>2</td></tr></table>		Carries blood of body	Carries blood to lungs	Carries blood from lungs	Carries blood from body	(a)	1	2	3	4	(b)	1	3	4	2	(c)	2	4	3	1	(d)	3	1	4	2	1
	Carries blood of body	Carries blood to lungs	Carries blood from lungs	Carries blood from body																							
(a)	1	2	3	4																							
(b)	1	3	4	2																							
(c)	2	4	3	1																							
(d)	3	1	4	2																							
2	The diagram given below represents the liver, kidney and some associated blood vessels. Identify the vessel from the labelled parts A-D in which the blood will contain the lowest concentration of urea.  (a) A (b) B (c) C (d) D	1																									

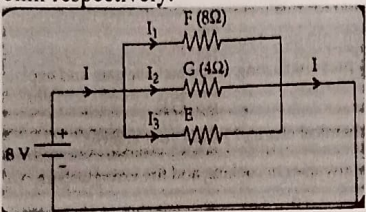
3	Which of the following comments applies to the brains of most animals? (a) Within the brain, neurons exchange information with one another. (b) Brains usually lie as near as possible to the important sensory structures in an animal. (c) Brains send action potentials to the hindmost portion of the animal by means of major nerves. (d) All of the above	1																				
4	In our body, calcium and phosphorus ions are controlled by: (a) thyroid gland (b) pituitary gland (c) adrenal gland (d) parathyroid gland	1																				
5	The diagram shows a section through the female reproductive system.  During pregnancy, where does mitosis occur in the cells of the embryo? <table><tr><td></td><td>X</td><td>Y</td><td>Z</td></tr><tr><td>(a)</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>(b)</td><td>✓</td><td>✓</td><td>×</td></tr><tr><td>(c)</td><td>✓</td><td>×</td><td>✓</td></tr><tr><td>(d)</td><td>×</td><td>×</td><td>✓</td></tr></table>		X	Y	Z	(a)	✓	✓	✓	(b)	✓	✓	×	(c)	✓	×	✓	(d)	×	×	✓	1
	X	Y	Z																			
(a)	✓	✓	✓																			
(b)	✓	✓	×																			
(c)	✓	×	✓																			
(d)	×	×	✓																			
6	A recessive homozygote is crossed with a heterozygote of the same gene. What will be the phenotype of the F₁-generation? (a) All dominant (b) 75% dominant, 25% recessive (c) 50% dominant, 50% recessive (d) 25% dominant, 50% heterozygous, 25% recessive	1																				
7	The diagram shows the flow of energy through an ecosystem.  The smallest amount of energy transferred between organisms and the largest amount of energy lost to the ecosystem is represented by which arrows? <table><tr><td></td><td>Smallest energy transfer</td><td>Largest energy loss</td></tr><tr><td>(a)</td><td>4</td><td>3</td></tr><tr><td>(b)</td><td>2</td><td>1</td></tr><tr><td>(c)</td><td>2</td><td>3</td></tr><tr><td>(d)</td><td>1</td><td>4</td></tr></table>		Smallest energy transfer	Largest energy loss	(a)	4	3	(b)	2	1	(c)	2	3	(d)	1	4						
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(d)	1	4																				
	The following two questions consist of two statements – Assertion (A) and Reason (R). Answer these questions by selecting the appropriate option given below: A. Both A and R are true, and R is the correct explanation of A. B. Both A and R are true, and R is not the correct explanation of A. C. A is true but R is false. D. A is false but R is true																					
8	Assertion : Mutation is sudden change in the genetic material. d	1																				

	Reason : Variation is useful for the survival of species over time.	
9	Assertion : Biomagnification is caused due to the accumulation of biodegradable toxicants in organisms at each successive trophic level. Reason : Biomagnification leads to the maximum accumulation of chemicals in small fishes.	1
10	Which digestive secretion does not contain any enzyme but is important? Discuss.	2,
11	How does feedback mechanism regulate hormone secretion? OR Tendrils encircle or coil around the object in contact with it. Elaborate.	2
12	Where are the genes located? What is the chemical nature of Gene?	2
13	An angiosperm plant having red coloured flowers when crossed with the other having the same colour produced 40 progenies, out of which 30 plants were with red coloured flowers, 10 plants were with white coloured flowers. Find out: (a) What is the possible genotype of parent plants? (b) Which trait is dominant and recessive? (c) What is this cross called as and what is its phenotypic ratio?	1 1 1
14	How is ozone formed in the upper atmosphere? Why is damage to the ozone layer a cause of concern to us? What causes this damage?	3
15	Thyroid gland is a bilobed structure situated in our neck region. It secretes a hormone called thyroxine. Iodine is necessary for the thyroid gland to make thyroxine. Thyroxine regulates carbohydrate, protein and fat metabolism in the body. It promotes growth of body tissues also.  When there is an excess of thyroxine in the body, a person suffers from hyperthyroidism and if this gland is underactive it results in hypothyroidism. Hyperthyroidism is diagnosed by blood tests that measure the levels of thyroxine and Thyroid Stimulating Hormone (TSH). Hypothyroidism is caused due to the deficiency of iodine in our diet resulting in a disease called goitre. (a) Where is the thyroid gland situated in our body? (b) State the function of thyroxine in the human body. (c) What is hyperthyroidism? How can we control hypothyroidism? OR Name the hormones secreted by Parathyroid and pancreas.	1 1 2 2
16	The reproductive parts of angiosperms are located in the flower. You have already studied the different parts of a flower – sepals, petals, stamens and pistils. Stamens and pistil are the reproductive parts of a flower which contain the germ-cells. The flower may be unisexual (papaya, watermelon) when it contains either stamens or pistil or bisexual (Hibiscus, mustard) when it contains both stamens and pistil. (a) What are the male reproductive parts of a flower?	1

	(b) What are the different parts of pistil? (c) Where does fertilization occur in flowers? OR What are the two types of pollination?	1 2 2
	SECTION -B	
17	Which of the following statements are correct? A. The chemical name of bleaching powder is calcium oxychloride. B. Bleaching powder is used for sterilization of drinking water. C. Bleaching powder is alkaline in nature hence it is used as an antacid. D. Bleaching powder is manufactured by Hasen Clever process. a) All of these b) A and C c) A, B and D d) B and C	1
18	Calcium oxide reacts vigorously with water to produce slaked lime. $\text{CaO(s)} + \text{H}_2\text{O(l)} \rightarrow \text{Ca(OH)}_2\text{(aq)}$ This reaction can be classified as : A. Combination reaction B. Exothermic reaction C. Endothermic reaction D. Oxidation reaction Which of the following is the correct option? a) C and D b) A and C c) A, C and D d) A and B	1
19.	The name and formula of third member of homologous series of alkyne is: a) Butyne C_4H_8 b) Propyne C_3H_6 c) Butyne C_4H_6 d) Propyne C_3H_4	1
20.	The property of metal by which it can be drawn into wires is called:  a) Ductility b) Conductivity c) Sonorous d) Malleability	1
21.	Which element is stored in water and which element is stored in kerosene? a) Calcium and phosphorus respectively b) Phosphorus and sodium respectively c) Sodium and phosphorus respectively d) Magnesium and sodium respectively	1
22.	Which of the following statements are correct regarding isomers? I. Isomers have the same structural formula. II. Isomers have the same molecular formula.	1

	III. Isomers have the same chemical properties. IV. The compound C_3H_8 has no isomers. a) II and IV only b) I, III and IV only c) I and III only d) I, II, III and IV	
23.	Bottle A contains oxalic acid and bottle B contains sodium carbonate solution. When pH paper is dipped in each of the solutions, the colour seen in A and B respectively be a) orange, blue b) blue, orange c) green, blue d) orange, green	1
24.	Assertion (A): The balancing of chemical equations is based on the law of conservation of mass. Reason (R): Total mass of reactants is equal to the total mass of products. a) Both A and R are true and R is the correct explanation of A. b) Both A and R are true but R is not the correct explanation of A. c) A is true but R is false. d) A is false but R is true.	1
25.	a. If you have phenolphthalein as an indicator, how will you test for acid and base? b. What will be the colour of a blue litmus paper on bringing it in contact with a drop of dil. NaOH?	2
26.	Silver chloride kept in a china dish turns grey in sunlight.  a. Write the colour of silver chloride when it was kept in the china dish. b. Name the type of chemical reaction taking place and write the chemical equation for the reaction. c. State one use of the reaction. Name one more chemical which can be used for the same purpose.	3
27.	 Anjali, a science student, decides to investigate the properties of sulphur. She takes a small amount of yellow sulphur powder in a hard glass test tube. She sets up the test tube with a clamp and gently heats it over a Bunsen burner flame. After the sulphur melts and begins to burn, she collects the gas evolved by inverting another test tube over the mouth of the first one. She then conducts two separate tests on the collected gas: Pratyush took sulphur powder on a spatula and heated it. He collected the gas evolved by inverting a test tube over it. What will be the action of gas on i. Dry litmus paper? ii. Moist litmus paper? Write a balanced chemical equation for the reaction taking place.	3
28.	Read the following text carefully and answer the questions that follow:	1+1+2

	Salts play a very important role in our daily life. Sodium chloride which is known as common salt is used in almost every kitchen. Baking soda is also a salt used in faster cooking as well as in the baking industry. The family of salts is classified on the basis of cations and anions present in them. a. Identify the acid and base from which Sodium chloride is formed. b. Find the cation and the anion present in Calcium sulphate. c. "Sodium chloride and washing soda both belong to the same family of salts." Justify this statement. OR Define the term pH scale. Name the salt obtained by the reaction of Potassium hydroxide and Sulphuric acid and give the pH value of its aqueous solution.	1+1+2
29.	a. Define a decomposition reaction. b. How can we say that (I) electrolysis of water, and (II) blackening of silver bromide when exposed to sunlight, are decomposition reactions? Mention the type of energy involved in each case. c. The type of reactions in which calcium oxide is formed, and calcium hydroxide is formed are opposite reactions to each other. Justify this statement with the help of chemical equations. OR a. What is the thermit process? Where is this process used? Write a balanced chemical equation for the reaction involved. b. Where does the metal aluminium, used in the process, occur in the reactivity series of metals? c. Name the substances that are getting oxidised and reduced in the process.	5
	SECTION C	
30	Strength of magnetic field produced by a current carrying solenoid DOES NOT depend upon: (a) number of turns in the solenoid (b) direction of the current flowing through the solenoid. (c) radius of solenoid (d) material of core of the solenoid	1
31	When a 4 V battery is connected across an unknown resistor, there is a current of 100 mA in the circuit. The value of the resistance of the resistor is: (a) 4 ohm (b) 40 ohm (c) 400 ohm (d) 0.4 ohm	1
32	The following question consists of two statements - Assertion (A) and Reason (R). Answer the question by selecting the appropriate option given below: (a) Both A and R are true, and R is the correct explanation of A. (b) Both A and R are true, and R is not the correct explanation of A. (c) A is true but R is false. (d) A is false but R is true. Assertion (A): Iron filings scattered around a straight current carrying conductor in a plane perpendicular to the length of the conductor arrange themselves in concentric circles. Reason (R): Magnetic field has both magnitude and direction.	1
33	Define the term 'power of accommodation'. Write the modification in the curvature of the eye lens which enables us to see the nearby objects clearly? OR During evening study, Ankit notices that the letters in his book appear blurred when he holds it too close to his eyes, but appear clear when he moves it away to a certain minimum distance. (a) Name the minimum distance at which objects can be seen clearly without strain. (b) State its value for a normal human eye.	2

34	A person is suffering from both myopia and hypermetropia. (i) What kind of lenses can correct this defect? (ii) How are these lenses prepared?	2
35	Three resistors in a circuit are attached as shown here. The resistance of F and G are 8 ohm and 4 ohm respectively.  The resistance of E is unknown. These resistors are connected to a battery with a potential difference of 8 V. (a) What type of arrangement of resistors are shown in the figure? (b) What is the resistance of E if 0.2 A current flows through it? (c) What is the total current flowing in the circuit?	3
36	(a) Write the relationship between focal length and radius of curvature of a spherical mirror. (b) Why is the refractive index of a medium always greater than one? Draw the ray diagram for light traveling into glass from air. (c) A lens has a power of -3 D. Find the focal length of the lens and also describe the nature of the lens?	3
37	An object is placed at a distance of 42 cm from a concave lens of focal length 14 cm. (a) Use lens formula to find the distance of the image from the lens. (b) List four characteristics of the image (nature, position, size and erect/inverted) formed by the lens in this case (c) Draw a ray diagram to justify your answer of part (b).	3
38	A person recently started having difficulty in reading books and seeing nearby objects clearly. He noticed that when he could see distant objects clearly from the bench he was sitting on, he had trouble focusing on his newspaper and phone screen. His eyes would feel strained after long periods of reading, and he often experienced mild headaches. Concerned, he visited an ophthalmologist. Following a detailed eye check-up, the doctor diagnosed him with hypermetropia (farsightedness). The doctor explained that in his eyes, the image of nearby objects was forming behind the retina instead of on it. This could be due to either a shortened eyeball or low converging power of the eye lens. To correct this defect, the doctor prescribed a lens of appropriate focal length to help him focus nearby objects clearly. After wearing the prescribed glasses, he could read comfortably again, and his eye strain was reduced significantly. (i) What type of lens should be used in his spectacles? (ii) His near point of eye is 42 cm. What should be the focal length of the lens he used ? (iii) What will be the power of the lens? OR (iv) Write one possible cause of this defect.	4
39	(a) What is an electromagnet? List any two uses. Draw a labelled diagram to show how an electromagnet is made. (b) What is the purpose of the soft iron core used in making an electromagnet? OR (a) What is the function of the earth wire in electrical instruments? (b) What is the usual current rating of the fuse wire in the line to feed lights and fans? (c) Is a fuse connected in series or in parallel in a household circuit? Explain. (d) What is a short circuit?	5